

Web School Project Management System

Project Architecture & Project Structure Documentation

Prepared for: Project Documentation / Portfolio / LinkedIn

Project Architecture

The application follows a layered Spring MVC architecture. The presentation layer handles the user interface, controllers process HTTP requests, services contain business logic, repositories communicate with the database, and JPA entities represent persistent data. Spring Security provides authentication and role-based authorization.

Layer	Responsibility
User Interface	Thymeleaf templates, HTML5, CSS3, JavaScript and Bootstrap provide the web interface.
Controller	Receives HTTP requests, validates request flow, invokes services and returns views/responses.
Service	Contains application and business logic and coordinates repository operations.
Repository	Uses Spring Data JPA to perform database operations.
Entity	JPA/Hibernate entity classes map Java objects to PostgreSQL tables.
Database	PostgreSQL stores application data persistently.
Security	Spring Security manages authentication, password encoding, roles and protected URLs.

Architecture Flow

USER / ADMIN

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THYMELEAF + HTML + CSS + JAVASCRIPT + BOOTSTRAP

|

SPRING MVC CONTROLLERS

|

SERVICE LAYER

|

SPRING DATA JPA REPOSITORIES

|

JPA / HIBERNATE ORM

|

POSTGRESQL DATABASE

SPRING SECURITY

|

+ ----- Authentication

+ ---- BCrypt Password Encoding

+ ---- Role-Based Authorization

+ ---- Protected /user/** and /admin/** resources

Project Structure

The project is organized into clear packages and resources so that presentation, business logic, persistence, security and domain models remain separated and maintainable.

```
Web_SchoolProject_Management/
■
■ ■ ■ src/
■   ■ ■ ■ main/
■       ■ ■ ■ java/
■           ■ ■ ■ com.school.schoolproject/
■               ■ ■ ■ config/
■               ■ ■ ■ controller/
■               ■ ■ ■ entities/
■               ■ ■ ■ handler/
■               ■ ■ ■ repository/
■               ■ ■ ■ service/
■           ■
■       ■ ■ ■ resources/
■           ■ ■ ■ templates/
■           ■ ■ ■ application.properties
■
■ ■ ■ pom.xml
■ ■ ■ README.md
■ ■ ■ mvnw
■ ■ ■ mvnw.cmd
■ ■ ■ .gitignore
```

Package Responsibilities

Package	Purpose
config	Application configuration and Spring Security configuration.
controller	HTTP endpoints and MVC request handling for admin and user modules.
entities	JPA entity classes representing school management data.
handler	Authentication-related handlers such as successful-login redirection.
repository	Spring Data JPA repositories for persistence and querying.
service	Business/application logic between controllers and repositories.
templates	Thymeleaf HTML views for dashboards, forms and user-facing pages.

Main Functional Modules

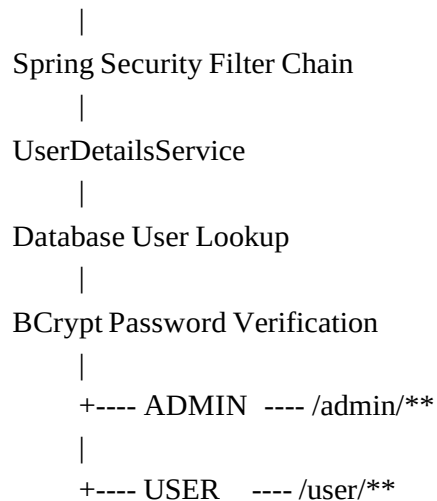
Module	Purpose
Student Management	Add, view, update, delete and manage student records.
Teacher Management	Manage teacher profiles, qualifications and employment information.
Admission Management	Handle admission records and related student/parent information.
Module	Purpose
Contact Management	Store and manage submitted contact information.
Enquiry Management	Manage school enquiries and enquiry messages.
Marksheet Management	Maintain student marksheet metadata and uploaded file information.
Dashboard	Provide centralized administration and quick access to management modules.
Campus & Academic	Present school campus facilities and academic information.

Technology Stack

Technology	Role
Java 17	Primary programming language
Spring Boot	Backend application framework
Spring MVC	Web/MVC layer
Spring Security	Authentication and authorization
Spring Data JPA	Data access abstraction
Hibernate	Object-relational mapping
PostgreSQL	Relational database
Thymeleaf	Server-side HTML rendering
HTML / CSS / JavaScript	Frontend development
Bootstrap	Responsive UI components and layout
Maven	Dependency and build management
Lombok	Reduction of Java boilerplate code

Security Flow

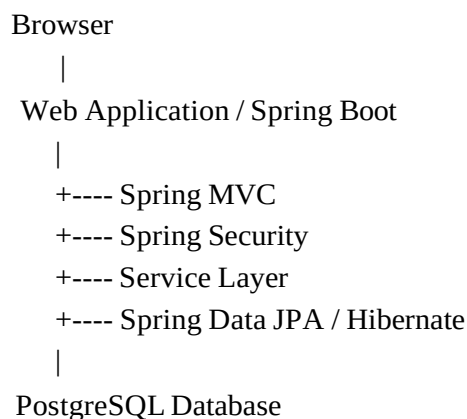
Login Request



Database & Persistence Flow

The application uses JPA entities as the domain model. Spring Data JPA repositories provide database operations, while Hibernate handles object-relational mapping between Java entities and PostgreSQL tables. Entity relationships are represented using JPA relationship annotations where required.

Deployment-Ready Architecture



Documentation Note: This document is based on the uploaded Web School Project Management System project structure. Before publishing, verify the final Java compiler target and database configuration in pom.xml and [application.properties](#) so the documentation exactly matches the deployed build.